Blockchain technology Unit-2

COMPUTER SCIENCE ENGINEERING (Jawaharlal Nehru Technological University, Kakinada)

HASHING :-

[Blockchain technology](#)
Blockchain technology is an intricate web of several technological innovations working together. Among the most important pieces of the blockchain puzzle is **hashing**.

Hashing is a cryptographic function that converts a string of characters of any length into a unique **output**, or **hash**, of fixed length. This means that no matter what combination of symbols are used as the **input**, they will always produce a one-of-a-kind **string of digits and characters**.

A Bitcoin hash looks like this:

00000000000000000025e2ba026a8ad462b9a693d80fd0887def167f5f888a11

(hash of block 540807)

HASHING ESSENTIALS

- Hashing is a method for cryptographically encoding data.
- It produces a fixed-length output from any input.
- The same input always produces the same hash.
- The input cannot be reconstructed from the hash.
- Modern hash functions make it virtually impossible to produce the same output from two different inputs.

Hashing in cryptocurrencies

Hashing is an integral part of all blockchain-based transactions, including the trading of cryptocurrency. Hash functions are necessary in everything from mining blocks to signing transactions to generating private keys.

Bitcoin	Bitcoin Cash	Ethereum	Litecoin	Ripple
SHA-256	SHA-256	keccak256	scrypt	SHA-512

A hash function is a mathematical algorithm used to calculate the hash. Different cryptocurrencies use different hash functions but all of them follow the same basic principles of hashing.

Main properties of hashing

Hashing produces outputs of fixed length

Hashing will always produce a unique, fixed-length output from any input. Let us take a look at what that means with a couple of examples.

Input	Output

hello	2CF24DBA5FB0A30E26E83B2AC5B9E29E1B161E5C1FA7425E73043362938B9
It’s a good day to HODL.	6B89D5D4AD6A3364410DD9BAB95FD250EF4A663D9D3C47CBD7388535A59
The entire novel Bleak House by Charles Dickens	4F144CC612CA27E2DD6DFD6663F68BABC3B758D602B5102BF14E717E823E

In the table above, the SHA-256 hash function is used to generate the hashes of three different inputs. In all three cases, the hash is completely unique, but its length remains the same. SHA-256 generates hashes that are 256 bits long, usually represented as 64 symbols comprised of numbers 0–9 and letters A–F. No matter how short or how long the input is – be it a single word (hello) or even a whole novel (*Bleak House* by Charles Dickens) – the hash is fixed at 64 characters.

Hashing is deterministic

The same input will always produce the same output. If you use SHA-256 to generate a hash from “fun”, you will always get the output seen in the table below. Even changing one letter, however, will produce a completely different hash.

Input	Output
fun	00C4285274FCC5D6FBA2EE58DAF0D8C2B9B825B68D35D65D0E90A9BB333A51B5
sun	27756F050E14A1CB1C1EE867F0EACE9EA4D9FCB81B8BEE089469F1EBD5FD7B17

Hashing is a one-way function

It is **infeasible** to determine what the input was from any given output. That is to say, it is virtually impossible to reverse the hash function with contemporary technology. The only way to determine what the input was is trying out random strings until you find the right one. This method is known as **brute force**.

Using brute force to reverse the hash back to the original string is easier said than done. No computer in existence is powerful enough to find the solution in any reasonable amount of time, nor are we ever likely to build one that will. Even IBM Summit, currently the fastest computer in the world, capable of making several trillion calculations per second, would need many years and an astounding amount of electricity to find the answer for a single hash.

Hashing is resistant to collisions

A **collision** occurs when a hashing mechanism produces the same output for two different inputs. This is possible in theory for hashing, as the number of unique hashes is limited but the number of inputs is not. However, the probability of collisions is extremely small. Hashing is thus said to be *resistant*, but not *immune*, to collisions.

SHA-256, the algorithm used by Bitcoin, outputs hashes that are 256 bits long (a 256 digit-long string of 1s and 0s). This means there are a total of 2^{256} unique hashes that it can produce. As soon as the number of inputs is larger than the number of all possible outputs, let us say $2^{256}+1$, at least two of the inputs will have the same output – that’s a collision.

So does that mean hashing is exploitable? No, not at all. 2^{256} is an enormous number. In fact, enormous does not even begin to do it justice. Think about it this way: 2^{256} is very roughly equal to the number of atoms in the entire observable universe. The sheer size of this number means the likelihood of a collision occurring is utterly miniscule.

Hashing is fundamental for blockchain

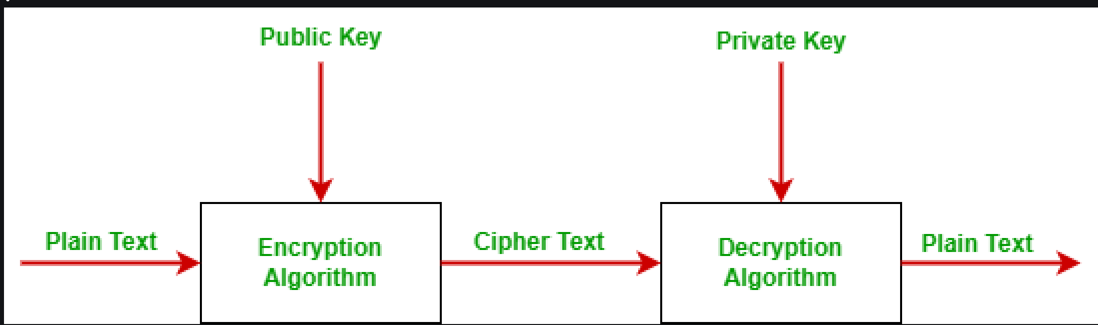
Blockchain technology combines a range of important solutions from various fields of science. The cryptographic hash function serves as the basis for building a blockchain and harnessing those solutions. It is difficult to imagine a decentralized network that forgoes hashing entirely.

PUBLIC KEY CRYPTO SYSTEM: -

Blockchain technology is one of the greatest innovations of the 21st century. In this article, we will focus on the concept of cryptography i.e. public-key cryptography or Asymmetric key cryptography.

Introduction To Public-Key Cryptography

Most of the time blockchain uses public-key cryptography, also known as asymmetric-key cryptography. Public key cryptography uses both public key and private key in order to encrypt and decrypt data. The public key can be distributed commonly but the private key can not be shared with anyone. It is commonly used for two users or two servers in a secure way.



Public Key: Public keys are designed to be public. They can be freely given to everyone or posted on the internet. By using the public key, one can encrypt the plain text message into the cipher text. It is also used to verify the sender authentication. In simple words, one can say that a public key is used for closing the lock.
Private Key: The private key is totally opposite of the public key. The private key is always kept secret and never shared. Using this key we decrypt cipher text messages into plain text. In simple words, one can say that the private key is used for opening the lock.

Why Do We Need Public-Key Cryptography?

- In symmetric-key cryptography, a single key is used to encrypt and decrypt the message. Here, the possibility of data loss or unauthorized access to data is high.

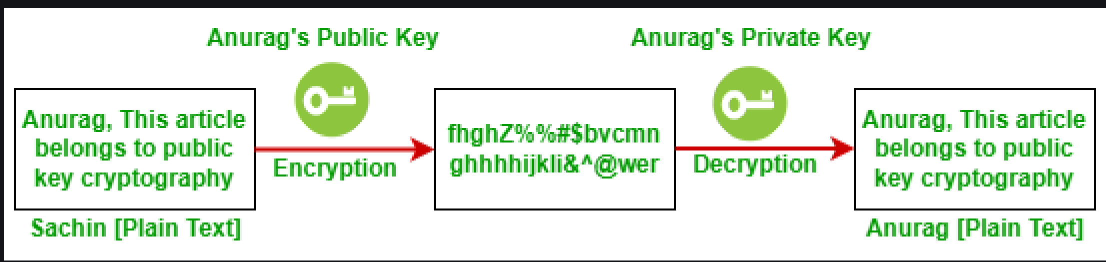
- To overcome the unauthorized access of data and data sent securely without any loss, we use public-key cryptography.
- Public-key cryptography is more secure than symmetric-key cryptography because the public key uses two keys to encrypt and decrypt the data
- Public-key cryptography allows users to hide the data that they want to send. The sender encrypts the data and the receiver decrypts the data. The encrypted message is not understood by unauthorized users.

Working On Public-Key Cryptography

Suppose, the sender wants to send some important message to the receiver.

- The sender first creates a message in the form of plain text which is in a readable format.
- The sender knows the public key of the receiver but doesn't know the private key of the receiver because the receiver keeps secret his private key. With the help of the public key of the receiver and the private key of the sender, the sender generates the encrypted message i.e. called cipher text. Cipher text is in an unreadable format. In this step, plain text converts into cipher text.
- Now, cipher text reaches the receiver end. The receiver knows its own private key, and with the help of the private key receiver converts the cipher text into readable format i.e. plain text.

The below example shows the working of public-key cryptography.



Let us try to under the working of public-key cryptography with an example. Suppose Sachin is the sender who wants to send a message to Anurag. Here Anurag is the receiver.

- Sachin uses Anurag's public key to encrypt the message and Anurag uses his own private key to decrypt the message.
- First Sachin creates plain text. Sachin has access to Anurag's private key and cipher text. Using Anurag's public key and his own public key,
- Sachin will generate an encrypted message i.e. cipher text which is in an unreadable format. After applying the encryption process plain text converts into cipher text.
- Now, Anurag receives a cipher text. First Anurag will decrypt the cipher text message into a readable format. For decrypting Anurag will use the private key. Now cipher text converts into plain text and is readable by the receiver. Because Sachin keeps his private key, Anurag knows that this message couldn't have come from anyone else. This is also called a digital signature.

Benefits of Public-key Cryptography

- **Authentication:** It ensures to the receiver that the data received has been sent by the only verified sender.
- **Data integrity:** It ensures that the information and program are changed only in a specific and authorized manner.
- **Data confidentiality:** It ensures that private message is not made available to an unauthorized user. It is referred to as privacy or secrecy.
- **Non-repudiation:** It is an assurance that the original creator of the data cannot deny the transmission of the said data to a third party.
- **Key management:** Public-key cryptography allows for secure key management, as the private keys are never transmitted or shared. This eliminates the need for a secure channel to transmit the private key, as is required in symmetric key cryptography.
- **Digital signatures:** Public-key cryptography allows for the creation of digital signatures, which provide non-repudiation and can be used to verify the authenticity and integrity of data.
- **Key exchange:** Public-key cryptography enables secure key exchange between two parties, without the need for a pre-shared secret key. This allows for secure communication even if the parties have never communicated before.
- **Secure communication:** Public-key cryptography enables secure communication over an insecure channel, such as the internet, by encrypting the data with the public key of the recipient, which can only be decrypted by the recipient's private key.
- **Versatility:** Public-key cryptography can be used for a variety of purposes, such as secure communication, digital signatures, and authentication, making it a versatile tool for securing data and communications.

Limitation of Public-Key Cryptography

- One can encrypt and decrypt the fixed size of messages or data. If there is an attempt to encrypt or decrypt a large size of the message then the algorithm demands high computational power.
- The main disadvantage of this algorithm is that if the receiver losses its private key then data/message will be lost forever.
- If someone has access private key then all data will be in the wrong hand.
- There are many secret-key which is faster than public-key cryptography.
- **Key distribution:** The process of securely distributing public keys to all authorized parties can be difficult and time-consuming, especially in large networks.
- **Performance:** Public-key cryptography is generally slower than symmetric-key cryptography due to its more complex algorithms, making it less suitable for applications that require fast processing speeds.
- **Security assumptions:** Public-key cryptography relies on mathematical assumptions about the difficulty of certain problems, such as factoring large numbers, which may not hold true in the future. As a result, public-key cryptography is vulnerable to future advancements in computing power and algorithmic breakthroughs.
- **Susceptibility to man-in-the-middle attacks:** Public-key cryptography is vulnerable to man-in-the-middle attacks where an attacker intercepts and alters

the public key before it reaches the intended recipient. This can result in the attacker being able to decrypt the message or impersonate the sender.

- **Complexity:** Public-key cryptography can be more complex to understand and implement than symmetric-key cryptography, requiring specialized knowledge and expertise.

- **Difference between Public and Private blockchain :**

S.no	Basis of Comparison	Public BlockChain	Private BlockChain
1.	Access –	In this type of blockchain anyone can read, write and participate in a blockchain. Hence, it is permissionless blockchain. It is public to everyone.	In this type of blockchain read and write is done upon invitation, hence it is a permissioned blockchain.
2.	Network Actors –	Don't know each other	Know each other
3.	Decentralized Vs Centralized –	A public blockchain is decentralized.	A private blockchain is more centralized.
4.	Order Of Magnitude –	The order of magnitude of a public blockchain is lesser than that of a private blockchain as it is lighter and provides transactional throughput.	The order of magnitude is more as compared to the public blockchain.
5.	Native Token –	Yes	Not necessary
6.	Speed –	Slow	Fast
7.	Transactions pre second –	Transactions per second are lesser in a public blockchain.	Transaction per second is more as compared to public blockchain.
8.	Security –	A public network is more secure due to decentralization and active participation. Due to the higher number of nodes in the network, it is nearly impossible for 'bad actors' to attack the system and gain control over the consensus network.	A private blockchain is more prone to hacks, risks, and data breaches/ manipulation. It is easy for bad actors to endanger the entire network. Hence, it is less secure.
9.	Energy Consumption –	A public blockchain consumes more energy than a private blockchain as it requires a significant amount of electrical resources to function and	Private blockchains consume a lot less energy and power.

		achieve network consensus.	
10.	Consensus algorithms –	Some are proof of work, proof of stake, proof of burn, proof of space etc.	Proof of Elapsed Time (PoET), Raft, and Istanbul BFT can be used only in case of private blockchains.
11.	Attacks –	In a public blockchain, no one knows who each validator is and this increases the risk of potential collision or a 51% attack (a group of miners which control more than 50% of the network's computing power.).	In a private blockchain, there is no chance of minor collision. Each validator is known and they have the suitable credentials to be a part of the network.
12.	Effects –	Potential to disrupt current business models through disintermediation. There is lower infrastructure cost. No need to maintain servers or system admins radically. Hence reducing the cost of creating and running decentralized application (dApps).	Reduces transaction cost and data redundancies and replace legacy systems, simplifying documents handling and getting rid of semi manual compliance mechanisms.
13.	Examples –	Bitcoin, Ethereum, Monero, Zcash, Dash, Litecoin, Stellar, Steemit etc.	R3 (Banks), EWF (Energy), B3i (Insurance), Corda.

HASH PUZZLES:-

An algorithm that transforms a given amount of data (the "message") into a fixed number of digits, known as the "hash," "digest" or "digital fingerprint." Hash functions are a fundamental component in digital signatures, password security, random number generation, message authentication and blockchains. A hash can also be used to avoid the risk of storing passwords on servers that could be compromised (see zero proof example). See RSA and hashing.

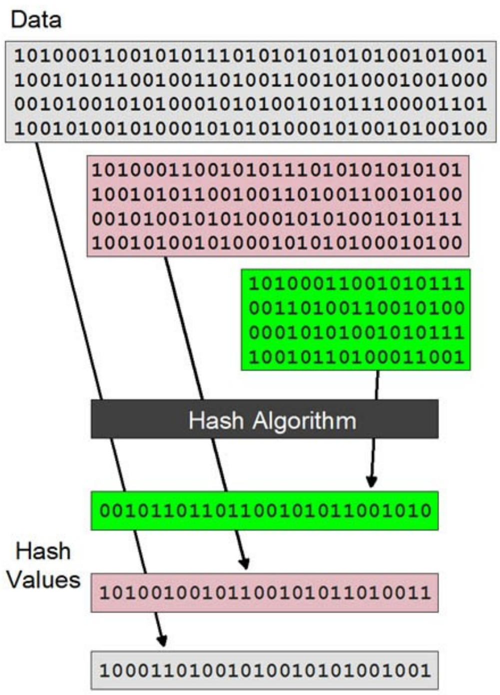
One-Way Processing
Also called a "one-way hash function" because it is nearly impossible to turn the digest back into the original data. It is also exceedingly rare that two different inputs can result in the same output.

Blockchain Integrity
This one-way processing guarantees that existing blockchain crypto balances and smart contract program code cannot be altered. All transactions in a block are hashed into the subsequent block creating a chained linkage, and any alteration to an existing block breaks the chain (for details, see blockchain). See Merkle tree, HMAC, digital

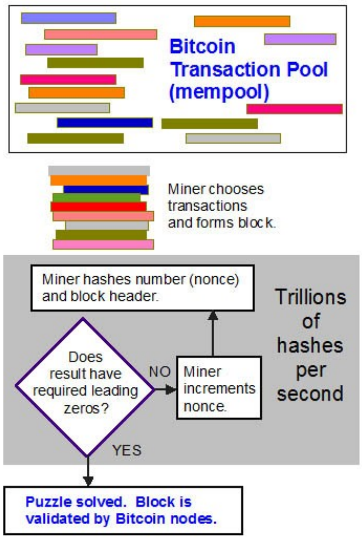
signature, MD5 and hash.

Solve the Bitcoin Puzzle by Hashing
As transactions are combined in a block by Bitcoin miners, they must solve a mathematical puzzle to prove they did work (see illustration below). The first to do so earns the right to add the new block to the blockchain and collect the fees and new coins. This proof-of-work (PoW) system is also used by Ethereum (see Ethash); however, that is changing (see Ethereum 2.0).

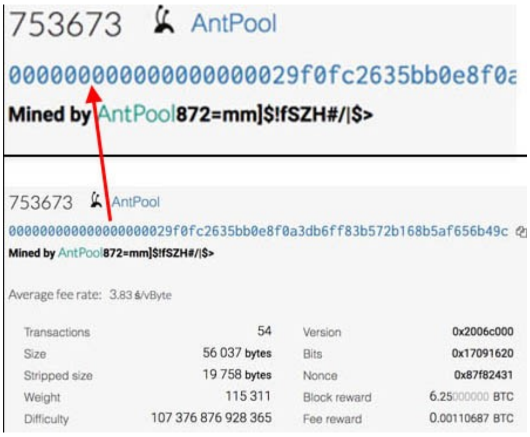
With specialized hardware executing trillions of hash computations per second, it can take a single machine years to come up with the required hash, which is why mining pools use hundreds of machines (see miner hardware). So much hash processing takes place mining Bitcoin worldwide that the electricity used could power a small country (see proof-of-work algorithm and hash rate). See crypto mining.



Hashes Are Fixed in Size
The hash value guarantees only that it is mathematically equal to the data it has hashed. If the data are changed in any way, that same hash cannot be generated. No matter how large or small the input, the hash output is fixed; for example, Bitcoin hashes are 256 bits long. See SHA.



Solving the Bitcoin Puzzle
The puzzle is finding the random number that, when added to the block's header, generates a hash with some number of leading zeros. Trying a new number trillions of times per second, the miner algorithm attempts to find the one that generates the desired results.



AntPool Wins Block 753673
The Bitaps website shows new Bitcoin blocks in real time. Notice the 19 leading zeros, which was the difficulty level at 17:38 UTC on 9-11-2022. The number of zeros is adjusted every 2,016 blocks to keep transactions flowing at one new block approximately every 10 minutes.

<u>EXTENSIBILITY OF BLOCK CHAIN CONCEPTS</u>:-

Extensibility refers to the ability of a system to adapt and evolve over time.

The **extensibility** of blockchain technology allows for the development of new use cases beyond its original intent. Extensibility is a critical factor in the ongoing success of blockchain technology.

Blockchain technology has a vast range of potential use cases, from supply chain management to identity verification to voting systems. The extensibility of blockchain technology allows for the creation of new use cases that were previously impossible. The potential use cases for blockchain technology continue to expand as the technology evolves.

It is necessary to understand the new ideas separately and together.

Blockchain Technology concepts include public-key and private-key cryptography, peer-to peer file sharing, distributed computing, network models, pseudonymity, blockchain ledgers, cryptocurrency protocols, and cryptocurrency.

It is a required to understand these concepts in order to operate in the blockchain technology environment. When you understand the concepts involved, it is not only possible to innovate blockchain-related solutions, but further, the concepts are portable to other contexts.

This extensibility of blockchain-related concepts may be the source of the greatest impact of blockchain technology as human agents understand these concepts and deploy them in every venue they can imagine.

One broad way of thinking about the use of blockchain concepts is applying them beyond the original context. The extensibility of blockchain technology allows for the creation of new use cases and the evolution of existing ones.

However, there are significant challenges to be addressed, including scalability, privacy, governance, and sustainability.

As blockchain technology continues to evolve, its potential use cases will continue to expand.

Some of the challenges to be addressed are

1. Smart Contracts
2. Interoperability
3. Scalability
4. Privacy
5. Governance
6. Sustainability

Smart Contracts:

Smart contracts are self-executing contracts with the terms of the agreement between buyer and seller being directly written into lines of code.

Smart contracts are a key innovation in blockchain technology, enabling the automation of complex processes.

Smart contracts have the potential to transform many industries, including financial services, real estate, and supply chain management.

Scalability

Scalability refers to the ability of a system to handle increasing amounts of work without impacting performance.

Scalability is a significant challenge for blockchain technology, which currently struggles with slow transaction processing and high fees.

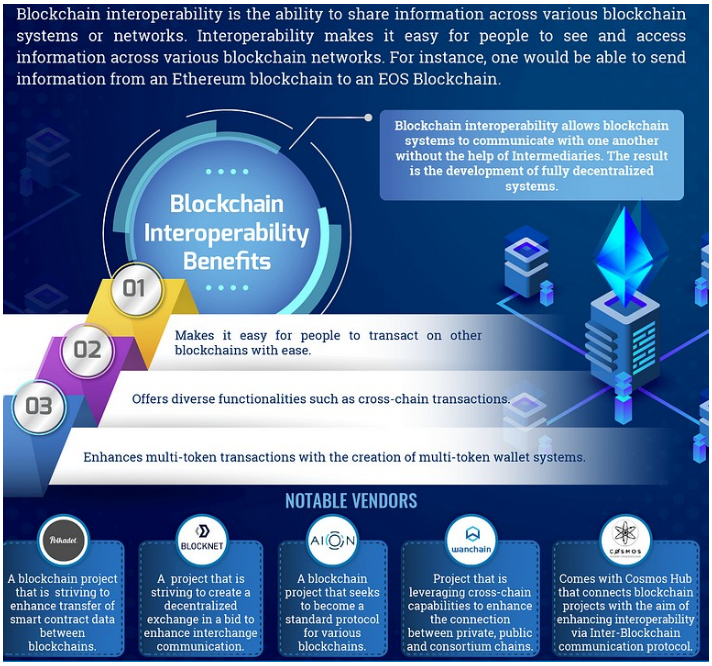
Several solutions are being developed to address scalability, including sharding, layer 2 solutions, and blockchain interoperability.

Interoperability

Interoperability refers to the ability of different blockchain networks to communicate and work together seamlessly.

Interoperability is critical for the widespread adoption of blockchain technology.

Several projects are underway to develop interoperability solutions, including Polkadot, Cosmos, and Ark.



Privacy

Privacy is a critical concern for many blockchain use cases, particularly those involving sensitive data.

Several privacy-focused blockchain projects, such as Monero and Zcash, have emerged to address this issue.

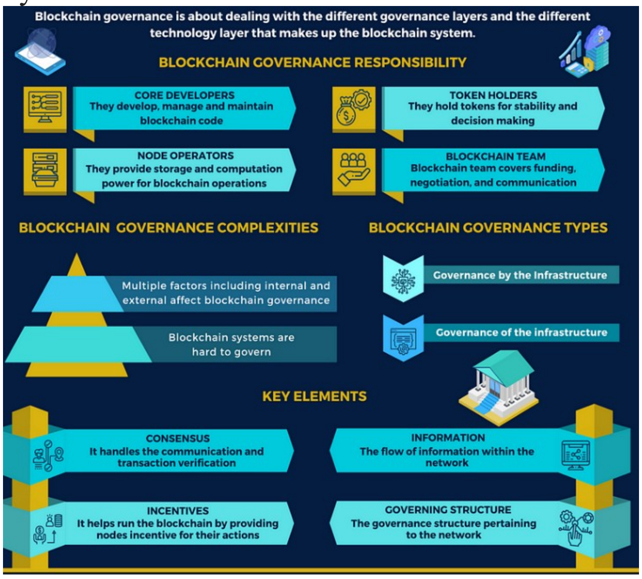
Privacy-enhancing technologies, such as zero-knowledge proofs, are also being developed to improve privacy on public blockchains.

Governance

Governance refers to the systems and processes in place to manage and regulate a blockchain network.

Effective governance is essential for the ongoing success of blockchain projects.

Several governance models exist, including on-chain governance, off-chain governance, and hybrid models.



Sustainability:

Sustainability refers to the ability of a blockchain network to operate over the long term.

Blockchain networks require significant computing power and energy consumption, which can be a barrier to sustainability.

Several projects are exploring alternative consensus mechanisms, such as proof of stake, to improve sustainability.

Some of the use cases are

1. Financial Services
2. Supply chain Management

3. Health Care
4. Identity Management
5. Voting Systems

Financial Services

Blockchain technology has the potential to transform the financial services industry by enabling faster, more secure, and more efficient transactions.

Use cases for blockchain in financial services include cross-border payments, trade finance, and digital identity.

Several blockchain projects are focused on financial services, including Ripple, Stellar, and Corda.

Supply Chain Management

Blockchain technology can improve supply chain management by increasing transparency and traceability, reducing fraud and counterfeiting, and improving efficiency.

Use cases for blockchain in supply chain management include tracking the origin and movement of goods, verifying product authenticity, and reducing waste.

Several blockchain projects are focused on supply chain management, including VeChain, Waltonchain, and Ambrosus.

DIGTIAL IDENTITY VERIFICATION:-

At Dock, we built our own digital identity blockchain because it was:

- **Faster:** Other blockchain options were congested and took a long time to finalize transactions
- **Most cost-effective:** There weren't many good solutions at the time for making transactions
- **Customization:** Our public, permissionless blockchain is built specifically for decentralized identity use cases to better accommodate customers
- **Precedence:** We're a first class application on our own chain in contrast to other chains where we would run the risk of being preempted by other work

The Dock blockchain serves as a foundation of trust by keeping an authentic record of all verifiable DIDs, public cryptography keys, and invalidation registries.

Verifiable Credentials that are issued are stored outside of the chain, usually in a holder's digital wallet app, along with its corresponding cryptographic key pairs. To ensure data privacy, the only data entered on the Dock chain are the issuer's and holder's DIDs, Credential Schema (its "template"), and Revocation Registries.

Digital Identity Blockchain Examples

These are just a few of many examples of how Dock's digital identity blockchain can be leveraged in various industries:

- **Secure identity verification:** Digital identity systems leveraging blockchain can instantly and securely verify the identity and credentials of individuals for various purposes, such as opening bank accounts, or accessing government services. Blockchain-based identity verification provides enhanced security as it eliminates the need for centralized third-party verification services and prevents identity fraud and theft.
- **Healthcare Records:** Patients can create and manage their own digital identity while healthcare providers can securely verify patient records and medical histories. This can result in providing better care while also ensuring data privacy and security.
- **Supply Chain Management:** Blockchain-based digital identities can be used to track and manage supply chain information, providing greater transparency and security. By creating digital identities for products, supply chain managers can track the movement of goods across the supply chain, ensuring product authenticity and preventing counterfeiting and fraud.

Digital Identity Blockchain Projects

These are just a few of a growing number of organizations using Dock's technology for digital identity blockchain projects:

- Gravity eliminates Health & Safety certificate fraud with Dock (South Africa)
- SEVENmile issues fraud-proof verifiable certificates using Dock (Australia)
- BurstIQ Makes Health Data Verifiable, Secure, and Portable With Dock (USA)

BLOCK CHAIN NEUTRALITY :-

Blockchain technology is transforming how markets work. Blockchains eliminate the need for trusted gatekeepers like banks to execute, verify, and record transactions. In the financial markets, their disruptive potential threatens both Wall Street banks and Silicon Valley venture capitalists. How blockchain technology is regulated will determine whether it encourages or inhibits competition. Some blockchain applications present serious fraud and systemic risks, complicating regulation. This Article explores the antitrust and competition policy challenges blockchain presents and proposes a regulatory strategy, modeled on Internet regulation and net neutrality principles, to unlock blockchain's competitive potential. It contends that financial regulators should promote blockchain competition—and the resulting market decentralization—except in cases where specific applications are shown to harm consumers or threaten systemic safety. Regulators also should ensure open access and non-discrimination on dominant blockchain networks. This approach will not only serve traditional antitrust goals of lowering prices and promoting innovation, but it also might achieve broader economic and social reform by reducing the power and influence of the biggest financial institutions.

DIGITAL ART :-

When talking about digital environments and art, the issue of authenticity and uniqueness of each work can arise. Creators and buyers of art and collectors of all kinds of digital assets have the same concerns: is this work authentic, is it unique, are there copies of a given digital work.

Digital art and millionaire collecting

And the sum of money that this new artistic current moves is not small. In 2021 the news broke: Beeple, a digital artist, became world famous when his work 'Everydays – The First 5000 Days' sold for 57 million euros. At the time, it became the third most expensive work of art ever auctioned by Christie's by a living artist.
Not all digital works have such an impact, but this milestone put this field on the map. The buyers of these pieces are consumers who are very familiar with cryptocurrencies and can start to get their hands on a collection of interesting pieces for much less than what Beeple's work cost. From around €600 it is possible to find interesting works, without the need to join more conventional and perhaps more elitist art and collecting circuits.

Precisely following this alliance of crypto-art and blockchain, in 2022, Telefónica launched a collection of 114 NFTs associated with a series of 114 unique drawings in digital format made by the chef Ferran Adrià, depicting the history of culinary evolution.

BLOCK CHAIN ENVIRONMENT :-

Blockchain technology has scaled rapidly in the last several years and is **projected to expand** to an even larger market across industries and countries. It is shaking up the global financial system, but environmental concerns risk stifling innovation. Potential regulatory action by governments concerned with the energy impact of blockchain activity is another reason to focus on sustainability.
In response, blockchain participants are considering environmental, social and governance (ESG) issues in their efforts to innovate. For example, Ethereum, the blockchain platform and the second largest cryptocurrency by market capitalization, Ether (ETH), is transitioning its consensus mechanism from proof-of-work to proof-of-stake in an effort known as "The Merge". One of the main drivers behind The Merge is the ability to measurably reduce energy consumption, indicating the urgency with which developers are factoring in their environmental impact.

To help navigate this complex landscape, PwC has developed a first-of-its-kind assessment framework that allows organizations to evaluate their environmental footprint as they look to take advantage of this emerging technology.

Breaking down blockchain technology

Blockchain is a distributed ledger technology that maintains data through a peer-to-peer network of computers. Although it was originally developed to facilitate cryptocurrency transactions, blockchain has expanded to other capabilities. From verifying ownership of **digital assets** like NFTs to tracking items in a supply chain, its versatility can drive innovation and support business transformation. Blockchain provides an opportunity for organizations to build trust through its decentralized, secure and transparent features, but its reputation as a threat to climate goals remains an obstacle.

The E in ESG: blockchain's environmental impact

A closer look reveals that not all blockchain protocols are the same and may have different purposes and varying levels of environmental impact.

A key component of these protocols is the consensus mechanism, which is the defined approach to validate transactions and prevent malicious activity. Each consensus mechanism has advantages and disadvantages regarding decentralization, security and scalability, and adjustments to these trade-offs can unlock opportunities to make the respective blockchain more sustainable.

A thorough analysis of blockchain protocols can inform the decisions of regulators, users and the market as this technology continues to grow. Blockchain has significant potential to **support sustainability**, and it may prove to be a valuable tool to help companies advance environmental aspects of their ESG goals.